Chapter 2

Applied Physics for CSE Stream

2.1 Module-1 : LASER and Optical Fibers

2.1.1 LASER

1. Expression for the number of photons emitted per t seconds $N = \frac{Pt\lambda}{hc}$ Photons.
   P is LASER Power Output in watt,
   t is the time in second,
   λ is the wavelength of LASER in m,
   h is Planck's Constant and
   c is the speed of light.

2. The Boltzmann relation $N_2 = N_1 e^{-\frac{hc}{\lambda kT}}$
   N_2 is the Number of Atoms in the higher energy state.
   N_1 is the Number of Atoms in the Lower Energy State,
   λ is the wavelength of LASER in m,
   k is Boltzmann Constant in JK^{-1},
   T is Absolute Temperature (in K.)

2.1.2 Optical Fibers

1. Expression for Numerical Aperture of an Optical Fiber $NA = \sqrt{\frac{n_1^2 - n_2^2}{n_0^2}}$
   n_0 is the RI of the surrounding medium,
   n_1 is the RI of the Core,
   n_2 is the RI of Cladding.

2. The Acceptance Angle, $\theta = Sin^{-1}(NA)$

3. Attenuation Co-efficient $\alpha = \frac{-10}{L} log_{10}\left(\frac{P_o}{P_i}\right)\ dB/km$
   L is the length of the fiber in km.
   P_o is the Power Output of the fiber in W.

P_i is the Power input of the fiber in W.
dB is the unit and stands for decibel.

2.2 Module-2 : Quantum Mechanics

1. The relation between Kinetic Energy and Momentum $E = \frac{p^2}{2m}$,
   m is the mass of the particle in kg,
   p is the momentum of the particle in Ns.

2. Energy of the photon $E = h\nu = \frac{hc}{\lambda}$,
   h is Planck's Constant in Js,
   ν is the frequency of the radiation in Hz,
   λ is the wavelength of the radiation in m,
   c is the speed of light ms^{-1}.

3. de Broglie Wavelength $\lambda = \frac{h}{p} = \frac{h}{mv}$ in $meter$
   h is Planck's Constant in Js,
   m is mass of the particle in kg,
   v is the velocity of the particle in ms^{-1}.

4. de Broglie Wavelength $\lambda = \frac{h}{\sqrt{2mE}}$
   h is Planck's Constant in Js,
   m is mass of the particle in kg,
   E is the Kinetic Energy of the particle in J.

5. de Broglie Wavelength $\lambda = \frac{h}{\sqrt{2mqV}}$
   h is Planck's Constant,
   m is the mass of the particle in kg,
   q is the charge on the particle in $coulomb$,
   V is the accelerating potential in $volt$.

6. de Broglie Wavelength $\lambda = \frac{h}{\sqrt{2m_e eV}} = \frac{12.27\times10^{-10}}{\sqrt{V}}\ m$
   h is Planck's Constant,
   m_e is the mass of the electron in kg,
   e is the charge on the electron in $coulomb$,
   V is the electron accelerating potential in $volt$.

7. Heisenberg's Uncertainty Principle
   $\Delta x \Delta p_x \geq \frac{h}{4\pi}$
   $\Delta E \Delta T \geq \frac{h}{4\pi}$
   Δx is the uncertainty in the measurement of Position in m,
   ΔP is the uncertainty in the measurement of Momentum in Ns,

ΔE is the uncertainty in the measurement of Energy in J.

ΔT is the uncertainty in the measurement of transition time in s.

8. The uncertainty in the measurement of momentum $\Delta P = m\Delta v$.
   Δv is the uncertainty in the measurement of velocity in ms^{-1}

9. Eigen Energy Values for a Particle in a one dimensional potential well of infinite depth
   $E_n = \frac{n^2h^2}{8ma^2}$,
   $n = 1, 2, 3...$ for the Ground, First excited and Second excited energy states etc.,
   h is Planck's Constant in Js,
   m is the mass of the particle in kg,
   a is the width of the potential well in m.

2.3 Module -3 : Quantum Computing

1. The wave function in Ket notation $|\psi\rangle$ (Ket Vector), ψ is the wave function.
   $|\psi\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$

2. The matrix for of the states $|0\rangle$ and $|1\rangle$.
   $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

3. Identity Operator $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

4. Pauli Matrices

   - $\sigma_0 = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.
   - $\sigma_1 = \sigma_x = X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.
   - $\sigma_2 = \sigma_y = Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$.
   - $\sigma_3 = \sigma_z = Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

5. A Matrix U is said to be Unitary Matrix if $U^\dagger U = I$,
   Here $U^\dagger$ is the conjugate-transpose of a matrix U.

6. A matrix A is Hermitian if $A^\dagger = A$

7. The wave function in Bra notation $\langle\psi|$ (Bra Vector), ψ is the wave function.
   $\langle\psi| = \begin{pmatrix}\alpha_1^* & \alpha_2^*\end{pmatrix}$
8. Inner Product $\langle\psi|\phi\rangle = \langle\psi| * |\phi\rangle$. Here $\langle\psi|$ is a Row Vector and $|\phi\rangle$ is a Column Vector. The result is always a scalar product.
9. The product $\langle\psi|\psi\rangle = |\psi|^2$, the probability density.
10. Orthogonality $\langle\psi|\phi\rangle = 0$

2.4 Module -4 : Electrical Properties of Materials and Applications

2.4.1 Electrical Conductivity in Solids

1. The Fermi Factor $f(E) = \frac{1}{e^{\left(\frac{E-E_F}{kT}\right)}+1}$
   E is the energy of the level above or below Fermi level in J,
   E_F is the Fermi Energy in J,
   k is Boltzmann Constant in JK^{-1},
   T is Absolute Temperature in K.

2.4.2 Superconductivity

1. The variation of Critical Field (Intensity) with Temperature $H_c = H_0\left[1 - \frac{T^2}{T_c^2}\right] Am^{-1}$,
   H_c is the critical field at a temperature T less than the critical temperature T_c in Am^{-1},
   H_0 is the critical field at $0K$ in Am^{-1}.
2. The variation of Critical Field (Flux Density) with Temperature $B_c = B_0\left[1 - \frac{T^2}{T_c^2}\right] tesla$,
   B_c is the critical field at a temperature T less than the critical temperature T_c in $tesla$,
   H_0 is the critical field at $0K$ in $tesla$.

2.5 Module -5 : Application of Physics in Computing

2.5.1 Physics of Animation

1. **The Odd Rule :**When acceleration is constant, one can use the Odd Rule to time the frames. With this method, one can calculate the distance the object moves between frames using a simple pattern of odd numbers. Between consecutive frames, the distance the object moves is a multiple of an odd number. For acceleration, the distance between frames increases by multiples of 1, 3, 5, 7,...
2. The Odd number multiplier for consecutive frames= $((frame\ number - 1) \times 2) - 1$
3. Multiplier for distance from first frame to current frame = $(current\ frame\ number - 1)^2$

4. $Base\ distance = \frac{Total\ distance}{(Last\ frame\ number-1)^2}$

5. $Jump\ Time = \sqrt{\frac{2 \times Jump\ Height}{g}}$ (for motion under earth's gravity)

6. Jump Magnification $JM = \frac{Jump\ time}{Push\ time} = \frac{Jump\ Height}{Push\ Height}$

7. $JM = \frac{Push\ Acceleration}{Jump\ Acceleration} = \frac{Push\ Acceleration}{Gravitational\ Acceleration}$

8. $\frac{Push\ Height}{Push\ Time} = \frac{Stop\ Height}{Stop\ Time}$

2.5.2 Statistical Physics for Computing

1. Poisson Distribution Probability Mass Function = $f(k;\lambda) = P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$
   Here λ is the mean number of events,
   k is the number of occurrences,
   e is Euler's Number (2.712),
   ! is the factorial.

2. The Decay Equation $N = N_0 e^{-\lambda t}$
   λ is decay constant s^{-1},
   t is the time in s,
   N_0 is Initial Number of Events,
   N is number of events after time t.